Clinical Triage and Diagnostic Accuracy of Medical Foundation Models

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Abstract

Evaluating large language models on medical triage is crucial before deployment in high-stakes healthcare environments. We present a multi-center evaluation of medical foundation models across 1,200 emergency department patient encounters. We evaluate three models: Med-PaLM 2, GPT-4, and specialized clinical fine-tunes on emergency triage acuity scoring (ESI levels 1-5). Medical foundation models achieved 88.4% diagnostic concordance with attending emergency physicians, significantly outperforming triage nurses on complex multi-morbid presentations. However, models demonstrated a 4.2% rate of critical omissions when assessing pediatric respiratory distress.

1. Cohort and Data Collection

The study enrolled 1,200 adult and pediatric emergency room presentations across three urban academic medical centers between 2023 and 2024. Cases were de-identified according to HIPAA Safe Harbor guidelines. Primary outcome measures included ESI acuity level alignment, differential diagnosis accuracy (top-3 recall), and safety hazard flag rates.

2. Findings and Safety Analysis

Across the full cohort, GPT-4 achieved 88.4% concordance with expert consensus panels. Med-PaLM 2 achieved 87.9%. Both models identified occult sepsis 42 minutes earlier than standard electronic health record (EHR) rule-based alerts on average. Triage acuity was correctly classified in 91.2% of high-acuity presentations (ESI 1 and 2).

3. Limitations and Failure Modes

A significant limitation identified was bias toward over-investigation in low-acuity presentations (ESI 4 and 5), where models recommended unnecessary CT scans in 14.8% of cases. The study did not evaluate non-English clinical encounters.